

preliminary
ENCODER

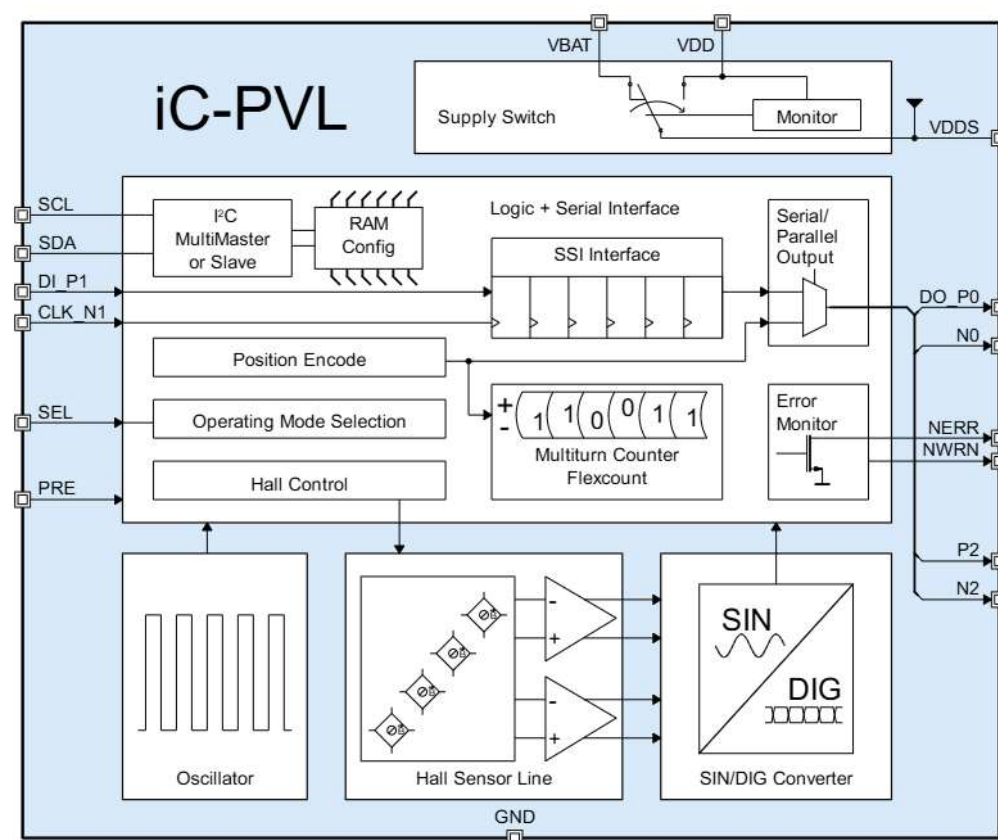
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APPLICATIONS

- ◆ Absolute hollow-shaft position encoders
- ◆ Gearless revolution counting
- ◆ Linear position sensors
- ◆ Metering applications
- ◆ Battery-powered portable equipment

QFN16
4 mm x 4 mm x 0.9 mm
RoHS compliant

BLOCK DIAGRAM

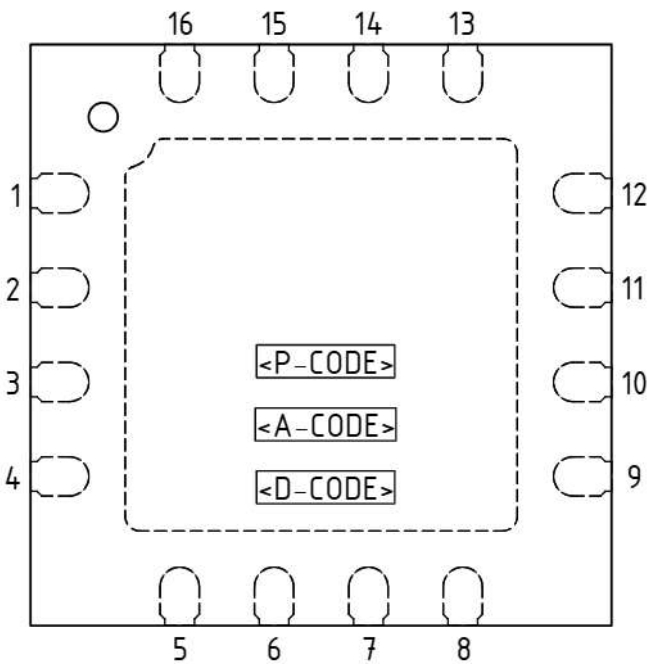


DESCRIPTION	
iC-PVL is an ultra low power magnetic encoder, used for linear and off-axis multiturn position sensing. On main power failure, iC-PVL switches automatically to battery supply and continues scanning the position. iC-PVL operates with pole wheels or linear scales with a pole width of 1.0 mm up to 5.0 mm. Due to various operating modes, iC-PVL can work with iC-Haus singleturn encoders (e.g. iC-MU, iC-MN, iC-LGC, iC-LNB, iC-MHM, etc.), as stand-alone SSI or incremental encoder, or links to embedded controllers via I²C. The Hall signal processing stage is designed for ultra low power applications and can be configured to support angular accelerations up to 180 000 rad/s² with 16 magnetic periods per revolution. The maximum magnetic signal frequency is 8 000 Hz. This corresponds to a rotational frequency of 30 000 rpm for magnetic scales with 16 pole pairs. With higher demands on acceleration, the power consumption increases. The maximum supported acceleration is configurable, therefore an optimal trade-off between power consumption and supported acceleration can be individually chosen to meet the demands of the target application. iC-PVL reads its configuration from an external EEPROM via an I²C interface with multimaster support. Among others, the bit length for multiturn and synchronization data, the interface mode, the maximum supported acceleration and the usage of error or par-	ity bits can be configured. The configuration read-in is triggered by the preset pin PRE. A pulse on this pin resets the device and reads a new configuration from the EEPROM. The multiturn counter is preset to a configurable value (default 0). The configuration RAM and multiturn counter value are protected against bit errors by an 8 bit CRC. Additionally, an error is generated on excessive speed or acceleration. An integrated battery monitor is used to signalize an empty battery as error. If an error is detected, it is displayed at output NERR and via the error bit in the SSI communication protocol. Additionally, a low battery voltage is indicated at output NWRN. Optionally, this warning can be transmitted in the serial data stream. Besides linear position encoding, the iC-PVL is also used for off-axis scanning of magnetic pole wheels. In these applications, a certain number of magnetic north-south field periods may be interpreted as one mechanical revolution. The FlexCount[®] circuitry offers this functionality. By electrically emulating the characteristics of a gear box, a transmission is freely programmable. For instance, 1-256 magnetic periods can be interpreted as one mechanical revolution. The iC-PVL multiturn encoder comes in a space-saving QFN16 package. This allows its integration in existing encoder systems or the design of smaller encoders.

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PACKAGING INFORMATION

PIN CONFIGURATION
QFN16 4 mm x 4 mm (top view)



No.	Name	Function
1	SEL	Mode Select Input ¹⁾ Low: Battery buffered counter with serial read-out High: 3 bit parallel complementary output Shorted to PRE input : I ² C slave mode
2	PRE	Preset Trigger Input
3	NERR	Error Output (active low)
4	SDA	I ² C Interface, Data Line
5	GND	Ground
6	VBAT	Battery Supply Voltage Input (typ. 3.6 V)
7	VDDS	Switched Supply Voltage Output
8	VDD	+3.0 V to 5.5 V Main Supply Voltage Input
9	N2	Parallel Output Bit 2 (negative logic), Incremental Output B
10	P2	Parallel Output Bit 2 (positive logic), Incremental Output A
11	N0	Parallel Output Bit 0 (negative logic)
12	NWRN	Battery Warning Output (active low)
13	DO_P0	Multiturn Interface, Data Output, Parallel Output Bit 0 (positive logic)
14	CLK_N1	Multiturn Interface, Clock Line, Parallel Output Bit 1 (negative logic)
15	DI_P1	Multiturn Interface, Data Input, Parallel Output Bit 1 (positive logic)
16	SCL BP	I ² C Interface, Clock Line Backside paddle ²⁾

IC top marking: <P-CODE> = product code, <A-CODE> = assembly code (subject to changes);
1) Do not leave pin open.
2) Connecting the backside paddle is recommended by a single link to GND. A current flow across the paddle is not permissible.

iC-PVL

LINEAR/OFF-AXIS

BATTERY-BUFFERED HALL MULTITURN ENCODER

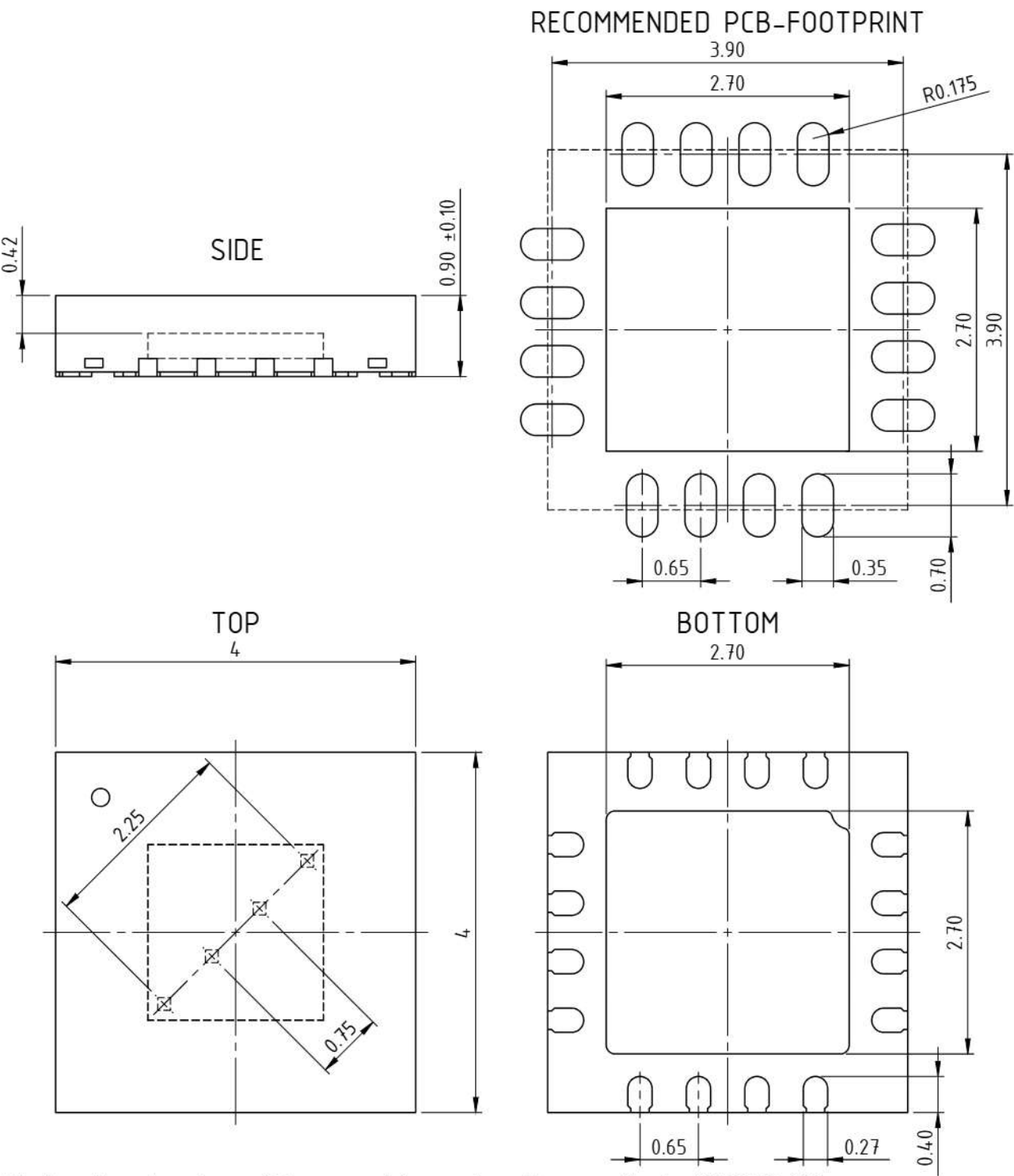
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PACKAGE DIMENSIONS QFN16 4x4



All dimensions given in mm. Tolerances of form and position according to JEDEC MO-220.
Positional tolerance of sensor pattern: ±0.10mm / ±1° (with respect to center of backside pad).

ABSOLUTE MAXIMUM RATINGS

These ratings do not imply operating conditions; functional operation is not guaranteed. Beyond these ratings device damage may occur.

Item No.	Symbol	Parameter	Conditions	Min.	Max.	Unit
G001	V(VDD)	Voltage at VDD		-0.3	6	V
G002	V(VBAT)	Voltage at VBAT	VDD > Von: VBAT < VDD + 1.0 V	-0.3	6	V
G003	V(VDDS)	Voltage at VDDS		-0.3	6	V
G004	V()	Voltage at SCL, SDA, DI_P1, CLK_N1, DO_P0, N0, NERR, NWRN, P2, N2, PRE, SEL		-0.3	6	V
G005	I(VDD)	Current in VDD		-10	50	mA
G006	I(VBAT)	Current in VBAT		-10	50	mA
G007	I(VDDS)	Current in VDDS		-10	50	mA
G008	I(GND)	Current in GND		-50	10	mA
G009	I()	Current in SCL, SDA, DI_P1, CLK_N1, DO_P0, N0, NERR, NWRN, P2, N2, PRE, SEL		-30	30	mA
G010	Vd()	ESD Susceptibility at All Pins	HBM, 100 pF discharged through 1.5 kΩ		2	kV
G011	Tj	Junction Temperature		-40	150	°C
G012	Ts	Storage Temperature Range		-40	150	°C

THERMAL DATA

Operating conditions:
VDD = 3.0...5.5 V, VBAT < VDD + 1.0 V

Item No.	Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
T01	Ta	Operating Ambient Temperature Range	package QFN16	-40		125	°C
T02	Rthja	Thermal Resistance Chip/Ambience	QFN16-4x4 surface mounted to PCB according to JEDEC 51 thermal measurement standards		40		K/W

All voltages are referenced to ground unless otherwise stated.
All currents flowing into the device pins are positive; all currents flowing out of the device pins are negative.

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ELECTRICAL CHARACTERISTICS

Operating conditions:
VDD = 3.0...5.5 V, VBAT < VDD + 1.0 V, Tj = -40...125 °C, fslow calibrated to 34 kHz with IBIAS, unless otherwise stated.

Item No.	Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
Total Device							
001	V(VDD)	Permissible Main Supply Voltage		3.0		5.5	V
002	I(VDD)	Supply Current in VDD	Tj = 27 °C, no load	1.5	4.0	6.0	mA
003	V(VBAT)	Permissible Battery Voltage	if VDD > Von: VBAT < VDD + 1 V	3.0	3.6	5.5	V
004	Iavg(VBAT)	Average Supply Current in VBAT	VBAT = 3.6 V, Tj = 27 °C, depending on fmag and A_MAX, see Table 34	1	10	800	µA
005	Ispike()	Peak Current in VDD and VBAT	tspike < 5 µs		4.0	10.0	mA
006	Vc()hi	Clamp Voltage hi at All Pins	Vc()hi = V() - VDDSD, I() = +1 mA	0.3	0.7	1.6	V
007	Vc()lo	Clamp Voltage lo at All Pins	I() = -1 mA	-1.6	-0.7	-0.3	V
008	tconfig	Power-Up Time After Preset	VDD > 3 V, initializing from EEPROM start address = 0x00, data valid start address = 0xA0, data valid		12 50	20 100	ms ms
009	C(VBAT)	External Bypass Capacitor at Pin VBAT	ceramic capacitor placed as close as possible to the pin	1			µF
Magnetic Signal Conditioning							
101	Hext	Permissible Magnetic Field Strength	at chip surface	10		100	kA/m
102	Bext	Permissible Magnetic Flux Density	at chip surface in air	12.5		125	mT
103	fmag	Magnetic Input Frequency	VDDSD = 3.0 V, tested via electrical input			8	kHz
104	frot	Permissible Rotation of Pole Wheel with	16 pole pairs 32 pole pairs 64 pole pairs			30000 15000 7500	rpm rpm rpm
105	vmax	Permissible Movement Speed (Linear)	1.5 mm pole width (3 mm magnetic period)			24	m/s
106	dsens	Diameter of Hall Sensor Circle	measured from center of each Hall plate, ONAX = 1		2.25		mm
107	hpac	Sensor-to-Package-Surface Distance	QFN16		0.4		mm
Oscillator Frequencies							
301	fslow	Slow Oscillator Frequency	calibrated to 34 kHz with IBIAS	32	34	36	kHz
302	ffast	Fast Oscillator Frequency	fslow calibrated with IBIAS	4.0	6.0	8.0	MHz
Supply and Battery Monitoring							
401	Von	Switch to VDD Supply (VDD Power On)	increasing voltage at VDD; VBAT > 3.0 V	2.8	2.9	3.0	V
402	Voff	Switch Back to Battery Supply (VDD Power Off)	decreasing voltage at VDD; VBAT > 3.0 V	2.7	2.8	2.9	V
403	Vhys	Hysteresis (VDD Switch)	Vhys = Von - Voff	25	100	150	mV
404	Vt()err	Battery Monitoring Error Threshold Voltage	BAT_THR = "11" BAT_THR = "10" BAT_THR = "01" BAT_THR = "00"	2.65 2.75 2.85 2.95	2.75 2.85 2.95 3.05	2.85 2.95 3.05 3.15	V V V V
405	Vt()wrn	Battery Monitoring Warning Threshold Voltage	BAT_THR = "11" BAT_THR = "10" BAT_THR = "01" BAT_THR = "00"	2.75 2.85 2.95 3.05	2.85 2.95 3.05 3.15	2.975 3.075 3.175 3.275	V V V V
406	Vew	Difference Battery Error-to-Warning	ΔVew = Vt()wrn - Vt()err	40	100	175	mV
Digital Outputs: DI_P1, CLK_N1, DO_P0, N0, P2, N2							
501	Vs()hi	Saturation Voltage hi	Vs()hi = VDDSD - V(), I() = -1.6 mA	0.05		0.4	V
502	Vs()lo	Saturation Voltage lo	I() = 1.6 mA	0.05		0.4	V
503	Isc()hi	Short-Circuit Current hi	VDDSD = 3.0 V, V() = GND	-15		-4	mA
504	Isc()lo	Short-Circuit Current lo	VDDSD = 3.0 V, V() = VDDSD	4		15	mA
505	tr()	Rise Time	CL = 30 pF			50	ns

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ELECTRICAL CHARACTERISTICS

Operating conditions:
VDD = 3.0...5.5 V, VBAT < VDD + 1.0 V, Tj = -40...125 °C, fslow calibrated to 34 kHz with IBIAS, unless otherwise stated.

Item No.	Symbol	Parameter	Conditions	Min.	Typ.	Max.	Unit
506	tf()	Fall Time	CL = 30 pF			50	ns
EEPROM Interface: SCL, SDA							
601	Vt()hi	Input Threshold Voltage hi			1.7	2	V
602	Vt()lo	Input Threshold Voltage lo		0.8	1.4		V
603	Vt()hys	Input Hysteresis	Vt()hys = Vt()hi - Vt()lo	75	200	500	mV
604	Vs()lo	Saturation Voltage lo	I() = 1.6 mA	0.05		0.4	V
605	Isc()lo	Short-Circuit Current lo	VDDS = 3.0 V, V() = VDDS	8		30	mA
606	Ipu()	Pull-Up Current	V() = 0 V...VDDS - 1 V	-1000	-300	-30	µA
607	fclk(SCL)	I ² C Output Frequency at SCL			f _{fast} / 128		kHz
Error Monitoring Output: NERR, NWRN							
701	Vs()lo	Saturation Voltage lo	I() = 1.6 mA	0.05		0.4	V
702	Isc()lo	Short-Circuit Current lo	VDDS = 3.0 V, V() = VDDS	4		15	mA
Digital Inputs: DI_P1, CLK_N1							
801	Vt()hi	Threshold Voltage hi			1.7	2	V
802	Vt()lo	Threshold Voltage lo		0.8	1.4		V
803	Vt()hys	Hysteresis	Vt()hys = Vt()hi - Vt()lo	75	200	500	mV
804	Ipd()	Pull-Down Current	V() = 1 V...VDDS	2	30	100	µA
Mode Select Input: SEL							
901	Vt()hi	Threshold Voltage hi			1.7	2	V
902	Vt()lo	Threshold Voltage lo		0.8	1.4		V
903	Vt()hys	Hysteresis	Vt()hys = Vt()hi - Vt()lo	75	200	500	mV
Preset Input: PRE							
A01	Vt()hi	Threshold Voltage hi			60	75	%VDD
A02	Vt()lo	Threshold Voltage lo		30	40		%VDD
A03	Vt()hys	Hysteresis	Vt()hys = Vt()hi - Vt()lo	0.7	1.0	1.4	V
A04	Ipd()	Pull-Down Current	V() = 1 V...VDDS, SEL = GND	10	120	300	µA
Serial Interface to Singleturn Sensor, SSI and Chain Mode							
B01	tp()	Propagation Delay: Clock Edge vs. DO Output		10		100	ns
B02	tout	Timeout	fslow calibrated via IBIAS	15	25	35	µs
Parallel Encoder Mode							
C01	tprocess	Processing Time (Parallel Out-put)	see Figure 11		10	30	µs

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OPERATING REQUIREMENTS: Serial and Parallel Interface

Operating conditions:
VDD = 3.0...5.5V, VBAT < VDD + 1.0V, Tj = -40...125 °C, fslow calibrated to 34 kHz with IBIAS, unless otherwise stated.

Item No.	Symbol	Parameter	Conditions	Min.	Max.	Unit
Serial Interface Mode (SSI or Chain Mode if Applicable)						
I001	t _{wait}	SSI Request after VDD Power-on		10		µs
I002	t _{req}	Request Signal lo Level Duration		50		ns
I003	t _c	Permissible Clock Period	due to Elec. Char. B02	0.25	2·t _{out}	µs
I004	f _c	Permissible Clock Frequency	due to Elec. Char. B02	1/(2·t _{out})	10	MHz
I005	t _{L1}	Clock Signal hi Level Duration		50		ns
I006	t _{L2}	Clock Signal lo Level Duration		50		ns
I007	t _{frame}	Cyclic Multiturn Data Frame Request Interval		85		µs
I008	t _s	Setup Time: Data stable before clock edge hi → lo	refer to Fig. 2	30		ns
I009	t _h	Hold Time: Data stable after clock edge hi → lo	refer to Fig. 2	30		ns
I ² C Slave Mode Direct Access						
I010	t _{wait}	I ² C Request after VDD Power-on		10		µs
I011	f _{scI}	Permissible Input Clock Frequency	I ² C standard mode with timeout Elec. Char. B02	1/(2·t _{out})	100	kHz
Parallel Encoder Mode (SEL = High)						
I012	t _{start}	Length of Start Pulse on PRE Pin	see Figure 11 on Page 21	1		µs
I013	t _{cycle}	Time Between Two Consecutive Sensor Read Cycles	see Figure 11 on Page 21	30		µs

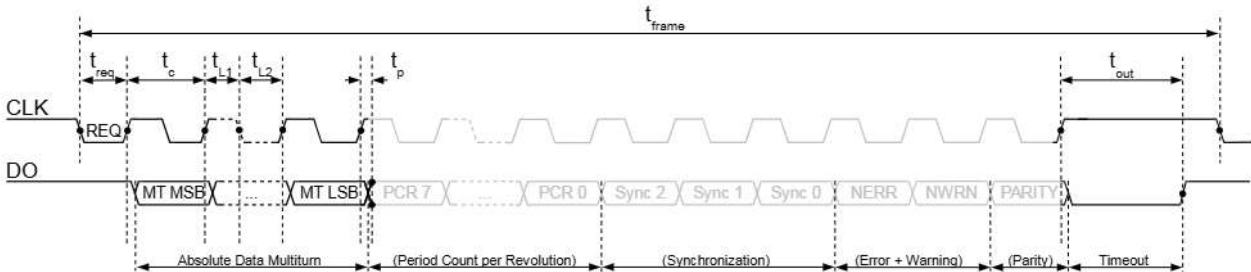


Figure 1: I/O line signals of the serial interface in SSI mode (INT_MODE = 0)

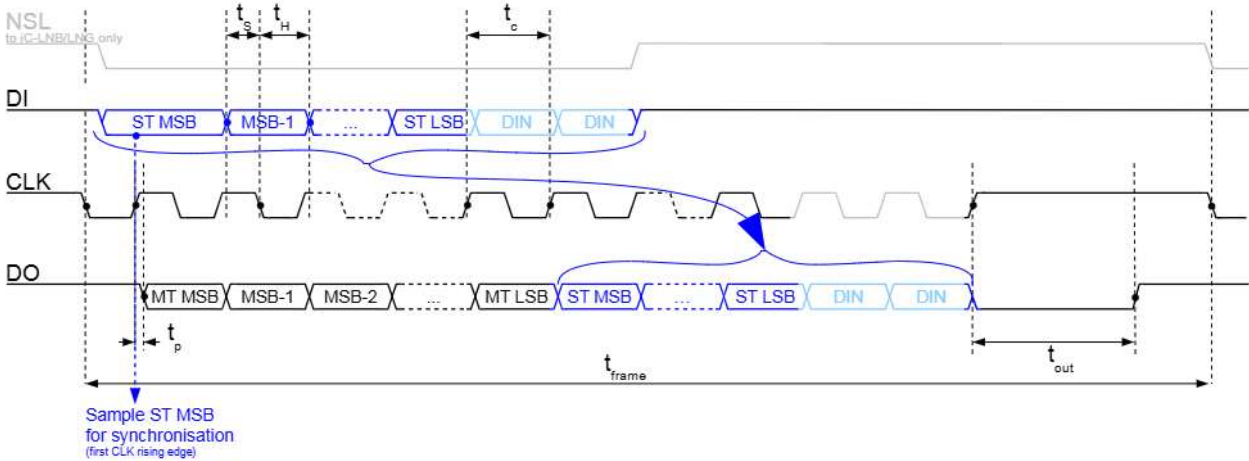


Figure 2: I/O line signals of the serial interface in chain mode (INT_MODE = 1).

CONFIGURATION PARAMETERS

Hall Sensor Signal Conditioning		Battery Monitor	
DIR:	Code Direction (P. 18)	BAT_MON:	Battery Monitoring Enable (P. 25)
OS:	Electrical Offset Multiturn to Singleturn (P. 18)	EN_WRN:	Low Battery Warning Enable (P. 20)
		BAT_THR:	Battery Monitor Thresholds (P. 25)
Serial Interface		Miscellaneous	
INT_MODE:	Serial Interface Operating Mode (P. 19)	POLEWID:	Pole Size of Magnetic Scale (P. 16)
MT_GRAY:	Multiturn Output Data Format (P. 19)	ONAX:	On-Axis Magnetic Scanning (P. 17)
ST_GRAY:	Singleturn Input Data Format (P. 19)	ABQUAD:	AB Quadrature Output (P. 20)
MT_BW:	Bit Width of Multiturn Data and Counter (P. 19)	HYS:	Hysteresis (P. 20)
SYNC_BW:	Synchronization Bit Width (P. 20)	I2C_POS:	Enable I ² C Position Read-out (P. 23)
PCR:	Period Count per Revolution (P. 17)	MT_PREL:	Multiturn Counter Preload Value (P. 21)
PCR_OUT:	PCR Output Mode (P. 17)	CHIP_REL:	Chip Release (P. 23)
EN_ERR:	Error Bit Transmission Enable (P. 20)	CRC Checksums	
EN_PAR:	Parity Bit Transmission Enable (P. 20)	CRC_CFG:	Checksum for Chip Configuration (0x00-0x05) (P. 22)
Bias and Oscillators		CRC_CTR:	Checksum for MT_PREL (0x07-0x0B) (P. 22)
IBIAS:	Bias Current; Oscillator Frequency Calibration (P. 27)		
A_MAX:	Maximum Angle Acceleration (P. 26)		

REGISTER MAP (EEPROM)

OVERVIEW								
Addr	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Interface and Hall Signal Processing								
0x00	EN_PAR		EN_ERR		DIR	ST_GRAY	MT_GRAY	INT_MODE
0x01	OS			MT_BW				
0x02	PCR							
Battery Monitor, Bias and Oscillators								
0x03	EN_WRN	BAT_MON	A_MAX			IBIAS		
Miscellaneous								
0x04	0	0	0	0	0	ONAX	POLEWID	
0x05	I2C_POS	PCR_OUT	SYNC_BW		BAT_THR		HYS	ABQUAD
CRC Configuration (inverted)								
0x06	CRC_CFG(7:0)							
Multiturn Counter: Preload Value								
0x07	MT_PREL(7:0)							
0x08	MT_PREL(15:8)							
0x09	MT_PREL(23:16)							
0x0A	MT_PREL(31:24)							
0x0B	MT_PREL(39:32)							
CRC Counter (inverted)								
0x0C	CRC_CTR(7:0)							

Table 5: Register map

REGISTER MAP (iC-PVL during I²C slave mode, ID = 0b1100 001)

OVERVIEW								
Addr	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Interface and Hall Signal Processing								
0x00	EN_PAR		EN_ERR		DIR	ST_GRAY	MT_GRAY	INT_MODE
0x01	OS			MT_BW				
0x02	PCR							
Battery Monitor, Bias and Oscillators								
0x03	EN_WRN	BAT_MON	A_MAX			IBIAS		
Miscellaneous								
0x04	0	0	0	0	0	ONAX	POLEWID	
0x05	I2C_POS	PCR_OUT	SYNC_BW		BAT_THR		HYS	ABQUAD
CRC Configuration (inverted)								
0x06	CRC_CFG(7:0)							
Multiturn Counter: Current Count (PCR_OUT= 0)					Multiturn Counter: Current Count (PCR_OUT= 1)			
0x07	MT_COUNT(7:0)				Current PCR(7:0)			
0x08	MT_COUNT(15:8)				MT_COUNT(7:0)			
0x09	MT_COUNT(23:16)				MT_COUNT(15:8)			
0x0A	MT_COUNT(31:24)				MT_COUNT(23:16)			
0x0B	MT_COUNT(39:32)				MT_COUNT(31:24)			
CRC Counter (inverted)								
0x0C	CRC_CTR(7:0)							
Synchronization Bits (I ² C slave mode read only)								
0x0D	0	0	0	0	0	SYNC(2:0)		
0x0E	Reserved							
Chip Release (I ² C slave mode read only)								
0x0F	CHIP_REL							
Status Register (I ² C slave mode only)								
0x10	PRESET	PDR	BAT_WRN	BAT_ERR	POS_ERR	CTR_ERR	CFG_ERR	STUP_ERR
Command Register (I ² C slave mode write only)								
0x11	CMD(7:0)							

Table 6: Register map during I²C slave mode

REGISTER MAP (iC-PVL during I²C slave mode, ID = 0b1100 000)

OVERVIEW								
Addr	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Status Register (I ² C slave mode only)								
0x00	PRESET	PDR	BAT_WRN	BAT_ERR	POS_ERR	CTR_ERR	CFG_ERR	STUP_ERR
Command Register (I ² C slave mode write only)								
0x01	CMD(7:0)							

Table 7: Register map during I²C slave mode

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OPERATING MODE SELECTION

The input SEL defines the operating mode of iC-PVL. For the default application as battery buffered multiturn counter with serial position read-out (SSI Interface), it is mandatory to connect the SEL input to GND potential. In case a three bit parallel and differential position output is desired, a high state at SEL input selects a parallel encoder mode. It is mandatory to connect SEL to a defined high or low potential.

An additional operating mode is activated when SEL is shorted to input PRE. In this mode, the preset pulse

does not trigger an EEPROM read-out. iC-PVL now behaves as I²C slave and all registers (configuration, position, status and command registers) are accessible via I²C device ID 0xC1. These are the first seven bits of the I²C addressing sequence, in binary representation 0b1100 001, this means group C = 0b1100, device 1 = 0b001.

The register layout visible via I²C is shown on page 11 for each particular address. Please note that some addresses are read only.

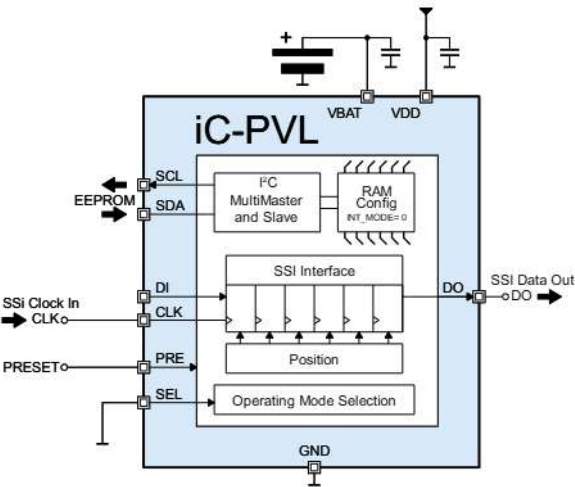


Figure 3: SEL = low: serial interface SSI mode

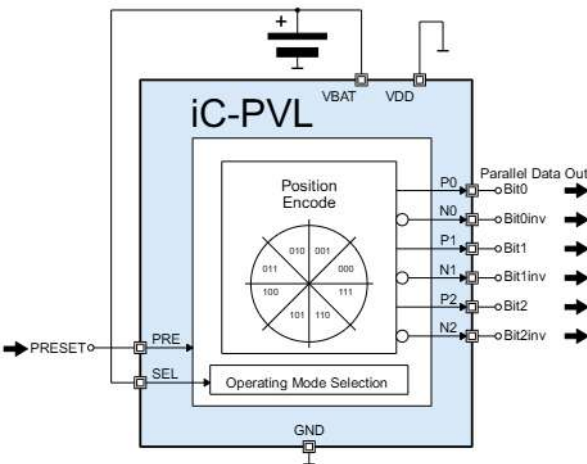


Figure 5: SEL = high: parallel encoder mode

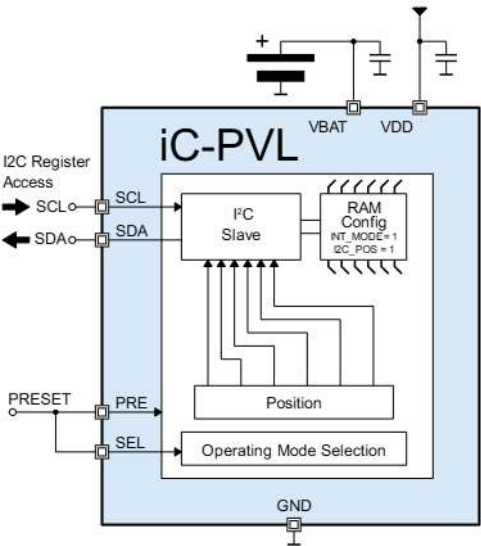


Figure 4: SEL = PRE: I²C slave mode

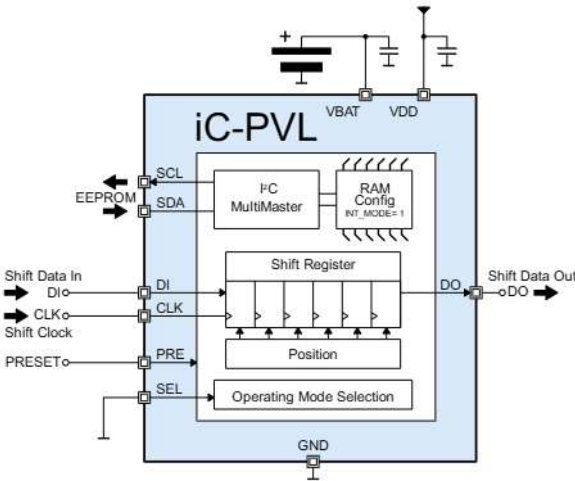


Figure 6: SEL = low: serial interface chain mode

OPERATING MODE SELECTION						
Operating Mode	Description	Pin SEL	Parameter INT_MODE (see Table 15)	Parameter I2C_POS (see Table 27)	I ² C Slave Function	I ² C Master Function (Read E2P cfg...)
SSI Mode (Fig. 3)	Battery buffered counter with serial read-out.	low	0	0	yes (without position read-out)	after preset pulse or I ² C command REBOOT
I ² C Slave mode (Fig. 4)	iC-PVL operates as I ² C slave, ID = 0b1100 001.	Shorted to PRE	0	1	yes (with position read-out)	only after I ² C command REBOOT
Parallel Mode (Fig. 5)	3 bit Parallel Encoder Mode, complementary output.	high	-	-	-	-
Chain Mode (Fig. 6)	Battery buffered counter with serial read-out.	low	1	0	no	only after preset pulse

Table 8: Operating mode selection, configuration settings, available functions

STARTUP BEHAVIOR

As shown in the previous chapter, iC-PVL can be booted up in four distinct operating modes. These are the SSI Mode, the Chain Mode, the Parallel Encoder Mode and the I²C Slave Mode, respectively. In case of a faulty startup procedure, an error is indicated at pin NERR.

Figure 7 shows the startup procedure of iC-PVL. The procedure starts when a battery supply is available. This would be the case if a battery is newly attached to the encoder system or the battery supply is switched on by an external microcontroller.

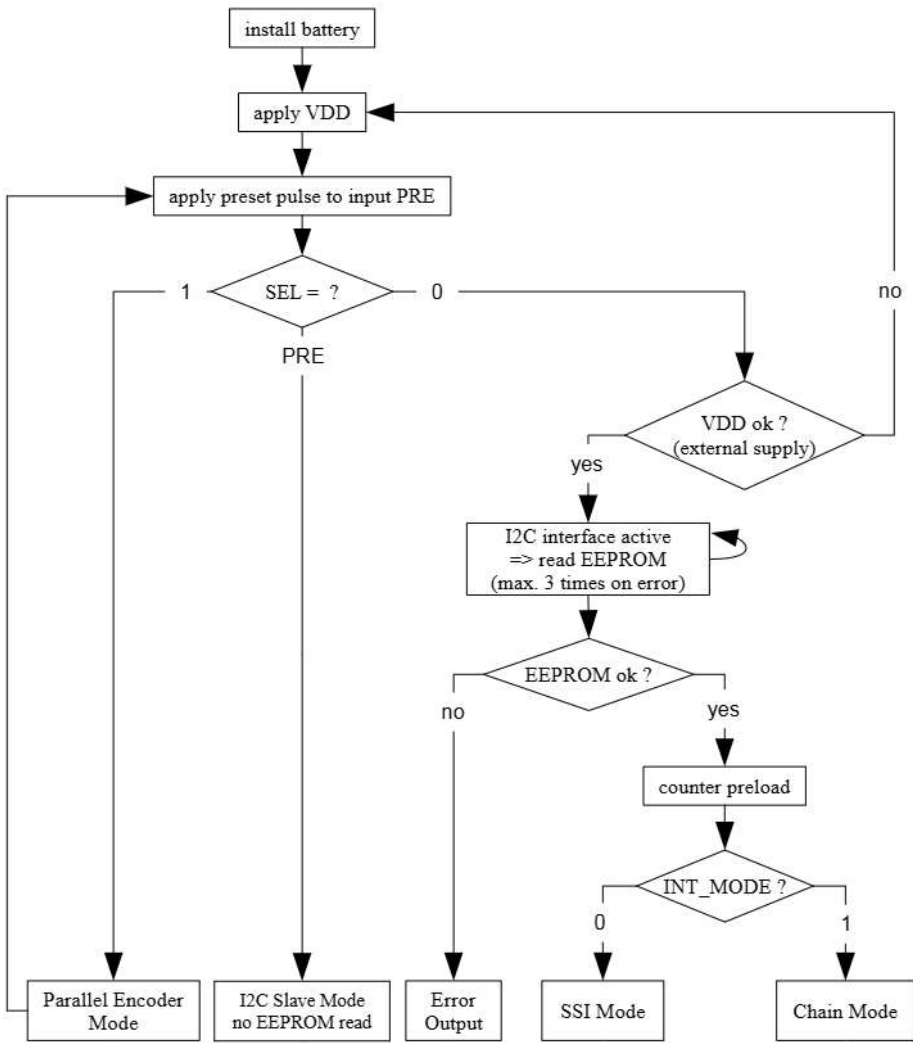


Figure 7: Startup behavior

Subsequent to powering up the VBAT pin, a suitable magnetic target needs to be installed, the main supply VDD powered, and a preset pulse (low-high-low) applied to input PRE. This resets the internal circuitry to its default initial state.

Via SEL, the operating mode is selected according to Table 8. If SEL = high, the chip functions as a 3 bit encoder with parallel complementary output and encodes the current position as shown in Figure 11.

In Parallel Encoder Mode, an external microcontroller activates the iC-PVL in distinct intervals to acquire the current position. The encoded position is valid when all complementary bits have changed their logic value (see Figure 11). After successful position read-out, iC-PVL is in ultra low power idle mode. The battery supply may then be switched off.

If SEL = low and for $V(VDD) > V_{on}$ during startup, the preset pulse triggers EEPROM read-out. Therefore, the VDD supply has to remain above V_{on} for at least t_{config} .

iC-PVL LINEAR/OFF-AXIS BATTERY-BUFFERED HALL MULTITURN ENCODER

preliminary



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The I²C multimaster tries to read the configuration data from an EEPROM connected to SCL and SDA. When the data is read without errors, the iC-PVL operates in SSI Mode or Chain Mode according to the configuration bit INT_MODE and stays in this operating mode as long as the battery supply is above V_{off} . After a third faulty attempt, EEPROM read-out is stopped and an error is indicated at pin NERR. Both serial interface modes are further explained on Page 19. I²C communication is possible in this operating mode, but position read-out via SSI or I²C is exclusive (refer to I²C SLAVE MODE on Page 23).

If SEL is shorted to input PRE, the device directly boots into I²C slave mode without any activity on the I²C lines. iC-PVL now responds to the I²C ID = 0b1100 001.

Note: After installation of the battery, it is important to put iC-PVL into a valid operating state to prevent the battery from an excessive current drain. To do so, a suitable magnetic target needs to be installed, the main supply VDD powered, and a preset pulse applied to input PRE.

Note: A magnetic field according to the specifications should be provided any time (see Elec. Char. No. 102). If a suitable magnetic target can not be provided during shipment or storage, use the sleep command (refer to Table 30) or disconnect the battery.

MAGNETIC SCALE SELECTION

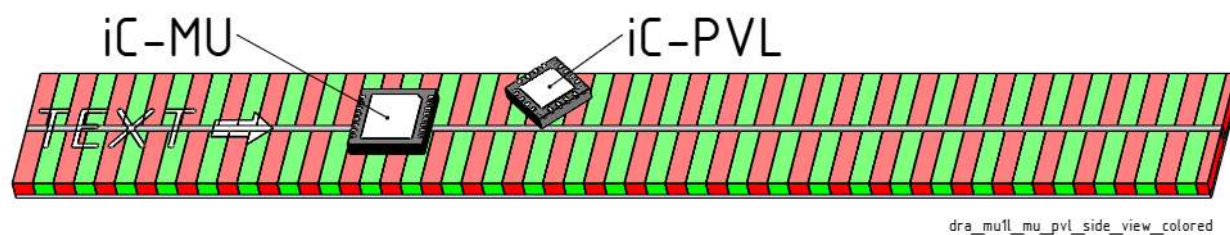


Figure 8: Positioning of iC-PVL on a linear scale (iC-MU nonius type)

iC-PVL is able to scan magnetic pole wheels or linear scales with a pole width of 1 mm to 5 mm. This corresponds to a magnetic period (N-S sequence) of 2 to 10 mm. Due to the diagonal sensor array, iC-PVL is rotated 45° in relation to the magnetic code (see Figures 8, 9, 10).

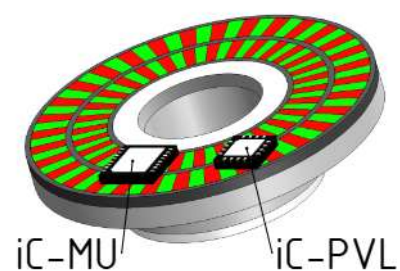


Figure 9: Positioning of iC-PVL on a pole wheel (iC-MU nonius type)

With the recommended differential scanning mode (POLEWID = 0x00), the measurement is highly tolerant to external common mode magnetic stray fields. The ideal pole width in this operating mode is 1.5 mm, but the device is functional with a pole width range of about 1 to 3 mm. This enables iC-PVL to use the same magnetic scale as a linear or off-axis singleturn sensor (e.g. iC-MU (1.28 mm), iC-MU150 (1.5 mm), iC-MHL200 (2 mm)).

POLEWID			
Addr. 0x04; bit 1:0			
Code	Pole size	Ideal size	Scanning
0x00	1 - 3 mm	1.5 mm	Off-axis, differential
0x01	4 - 5 mm	4.5 mm	Off-axis, single-ended
0x02	2 - 4 mm	3.0 mm	Off-axis, single-ended
0x03	1 - 3 mm	1.5 mm	Off-axis, single-ended

Table 9: Pole size of magnetic scale

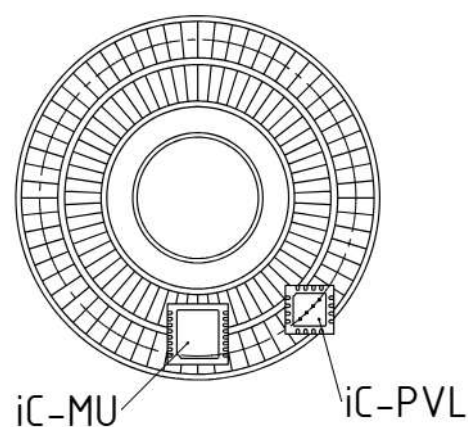


Figure 10: Positioning of iC-PVL on a pole wheel (iC-MU nonius type).

Together with the iC-MU off-axis nonius encoder, a magnetic multiturn encoder can be realized (see Figure 13, P. 28). The iC-PVL Hall sensor array is placed over the center line of a magnetic track, e.g. at a radius of 13.375 mm to scan the master track of the MU18S 30-32N magnetic target (see Figure 10). In this case, the corresponding PCR value for 32 magnetic periods ($PCR = 0x1F$) has to be applied (see Table 10).

In these applications, a certain number of magnetic periods (i.e., north-south pole pairs) may be interpreted as one mechanical revolution. The FlexCount® logic offers this functionality. By electrically emulating the characteristics of a gear box, the gear transmission is freely programmable. 1 to 256 pole pairs can be interpreted as one mechanical revolution. The configuration parameter PCR, period counts per (mechanical) revolution, is used as defined in Table 10.

PCR		Addr. 0x02; bit 7:0
Code	Period counts per revolution	
0x00	1	
0x01	2	
0x02	3	
...	Code + 1	
0xFE	255	
0xFF	256	

Table 10: Period counts per (mechanical) revolution

By way of example, assume a magnetic code disc with 32 periods. iC-PVL will make a multiturn count every 32 periods. The synchronization bits are distributed evenly over these 32 periods, i.e. the disc is divided in eight sectors. Therefore, one sector consists of 8 periods. While reading the position information, it looks like the iC-PVL would be placed above a normal diametrically polarized magnet. This scheme is also valid for non-binary, decimal or odd counts per revolution.

PCR_OUT		Addr. 0x05; bit 6
Code	Mode	
0	No output of PCR in serial data stream	
1	PCR is transmitted as 8 LSBs of MT data (Only for PCR > 0x00)	

Table 11: Period count per revolution output mode

Alternatively, the current magnetic period of the mechanical revolution can be transmitted in the serial data stream with option PCR_OUT = 1. In this mode, the eight LSBs of the multiturn data are used for PCR output. Unused bits are filled with zeroes. As a consequence, the multiturn counter is limited to 32 bits in this mode. The multiturn, PCR and synchronization information give the exact position of the code disc down to one eighth of a magnetic period.

Even as iC-PVL is particularly designed for off-axis scanning of magnetic pole wheels, it can also be used for on-axis scanning of a diametric cylindrical magnet. The on-axis scanning mode is activated with the configuration bit ONAX = 1. The sensor circle diameter is specified in Elec. Char. 106. In this mode, setting POLEWID to 0x01 is mandatory for differential scanning.

ONAX		Addr. 0x04; bit 2
Code	Magnetic scanning mode	
0	Off-axis (default)	
1	On-axis, differential (POLEWID = 0x01 required)	

Table 12: Use off- or on-axis magnetic scanning

COMPENSATION OF ASSEMBLY: Data Offset and Counting Direction

Assembled together with a magnetic code disc, code wheel or magnetic tape, the integrated Hall sensor signal processing generates a three bit position word, i.e. eight positions per magnetic period. Therefore, the iC-PVL provides up to three synchronization bits to the singleturn sensor in SSI read-out mode.

The position can be electrically manipulated to achieve the desired (leading or trailing) phase shift, regardless of the actual mounting position. This is useful if the phase relationship between an additional singleturn iC and the iC-PVL as multiturn encoder is unknown, or the singleturn sensor takes care of the synchronization (SSI mode) and expects a defined phase relationship. An offset value is added to the digitized Hall sensor position according to parameter OS (see Table 13).

In applications where the chain operation mode is used (see Page 19), the iC-PVL takes care of synchronization. Therefore, it has to be mounted in a leading position in relation to the singleturn iC. If the mounting position varies from that, the OS parameter can be used to achieve this phase shift. To ensure correct synchronization of multiturn and singleturn data, the resulting phase shift between the multiturn and singleturn position must be within the range of 0° to 180° (MT leading).

OS	
Addr. 0x01; bit 7:5	
Code	Phase shift
000	0°, no shift
001	+ 45° leading
010	+ 90° leading
011	+ 135° leading
100	± 180° leading or trailing
101	- 135° trailing
110	- 90° trailing
111	- 45° trailing

Table 13: Offset multiturn to singleturn

Note: 0° to 180° is the ideal range for tolerated values of phase shift between ST and MT. This range is further reduced due to communication, propagation or processing delays for the specific application. Typically, it is reduced by a few degrees, but increases with the signal frequency.

DIR	
Addr. 0x00; bit 3	
Code	Code direction
0	Normal
1	Inverted

Table 14: Code direction

The counting direction can be easily swapped with the configuration bit DIR. The bit would be typically used to invert the counting direction if the iC-PVL is assembled rotated or flipped.

SERIAL INTERFACE MODE (SEL = LOW)

iC-PVL can be connected to a singleturn sensor or interpolator via its serial interface. To ensure compatibility with different devices, iC-PVL's serial interface can operate in two distinct modes: standard SSI or in a chain mode.

In SSI mode, iC-PVL replies to a multiturn data request of an SSI protocol master. The master takes care of data synchronization and calculates a consistent absolute position. Data input DI is not used in this operating mode. iC-PVL is compatible with all iC-Haus optical or magnetic singleturn encoders featuring a multiturn interface (MTI). For details refer to the application examples of Figure 13 and Figure ?? on Page 28. In SSI mode, iC-PVL can also be used as stand-alone magnetic period counter. The position data is output in an SSI compatible protocol (see Figure 15 on Page 29).

In chain mode, a singleturn sensor transmits its singleturn position to the data input DI of iC-PVL. In this mode, iC-PVL takes care of synchronization, calculates a consistent absolute position and outputs at data output DO, comparable to a shift register, with MSB first.

The operating mode is set using parameter INT_MODE.

INT_MODE		Addr. 0x00; bit 0
Code	Mode	
0	Standard SSI read-out mode (SSI mode)	
1	Chain mode (chain mode)	

Table 15: Serial interface operating mode

By default, the iC-PVL expects binary data at input DI, and also outputs its counted multiturn position (and PCR) at DO in binary format. In case one data or both data is required in Gray format, this can be configured by setting the ST_GRAY bit or the MT_GRAY bit, respectively.

ST_GRAY		Addr. 0x00; bit 2
Code	Format	
0	Binary code	
1	Gray code	

Table 16: Singleturn input data format via port DI (in chain mode)

MT_GRAY		Addr. 0x00; bit 1
Code	Format	
0	Binary code	
1	Gray code	

Table 17: Multiturn and PCR output data format via port DO

Note: The usage of ST_GRAY = 1 in chain mode is limited. If activated, all data to input DI is converted to binary code. This conversion may lead to an unclear ending of the SSI communication. In this case, instead of a timeout at logic low level, a timeout at logic high level is received. If this is not appropriate, the exact singleturn input bit width can be defined by configuration parameter SYNC_BW which is not used in chain mode.

SYNC_BW		Addr. 0x05; bit 1:0
Code	Input bit width	
00	all bits (default)	
01	14 bit	
10	16 bit	
11	18 bit	

Table 18: Number of expected and converted input bits at pin DI with ST_GRAY = 1

The internal multiturn counter is 40 bit wide. In applications where smaller counter depths are sufficient or the bit width of the serial interface is limited, the output length of the counter value can be configured with MT_BW as shown in Table 19.

MT_BW		Addr. 0x01; bit 4:0
Code	Bit width	
0x00	9 bit	
0x01	10 bit	
...	...	
0x1E	39 bit	
0x1F	40 bit	

Table 19: Bit width of multiturn data and counter

Additionally, iC-PVL can transmit up to 3 synchronization bits, according to configuration parameter SYNC_BW shown in Table 20.

SYNC_BW Addr. 0x05; bit 1:0		
Code	Bit width	Tolerable phase shift range
00	0 bit	no synchronization bit
01	1 bit	0° ... 180°
10	2 bit	0° ... 270°
11	3 bit	0° ... 315°

Table 20: Synchronization bit width and resulting tolerable ideal phase shift

After the transmission of the absolute position and the synchronization information, iC-PVL's serial protocol allows the optional transmission of an error bit, a warning bit and a parity bit.

The error bit signalizes if a startup error, a wrong CRC checksum, an empty battery or position error (e.g. overspeed or magnet loss). Its polarity is configured with parameter EN_ERR shown in Table 21.

The warning bit represents an early battery warning. It indicates a low battery while the system is still functional. The polarity of the warning bit follows the polarity configured with EN_ERR. Details regarding error and status information are explained on Page 23. The optional parity bit finishes the transmission. Its polarity is either even or odd according to parameter EN_PAR.

Finally, the last 8 LSBs of the multiturn data can be used to transmit the period counts revolution in off-axis applications. See Table 11 for details.

The line signals for both interface modes are shown in Figures 1 and 2 on Page 9. Optional bits are greyed-out.

The number of transmitted multiturn and singleturn bits depends on parameter MT_BW and ST_BW.

EN_ERR Addr. 0x00; bit 5:4	
Code	Mode
00	Communication without error bit
01	Calibration mode
10	Communication with additional error bit (negative polarity)
11	Communication with additional error bit (positive polarity)

Table 21: Error bit enable

EN_WRN Addr. 0x03; bit 7	
Code	Mode
0	Communication without warning bit
1	Communication with additional warning bit (polarity as configured via EN_ERR)

Table 22: Warning bit enable

EN_PAR Addr. 0x00; bit 7:6	
Code	Mode
00	Communication without parity bit
01	reserved
10	Communication with additional parity bit (even polarity)
11	Communication with additional parity bit (odd polarity)

Table 23: Parity bit enable

OUTPUTS N0, P2, N2

In serial interface (P.) or I²C slave mode (P.), the outputs N0, P2, N2 provide either parallel or incremental position information. A magnetic period is divided in eight sectors. By default, these three bits are output in real-time at N0, P2, N2.

ABQUAD Addr. 0x05; bit 0				
C.	Mode	N2	P2	N0
0	Parallel position	MSB	MSB-1	LSB
1	Quadrature AB	B	A	none
N.	The output is inverted to the internally generated position.			

Table 24: AB quadrature output

With configuration parameter ABQUAD, a quadrature output can be activated. In this mode, A and B are

output at P2 and N2. One magnetic period is interpolated by a factor of two, i.e., two quadrature periods are observed per period corresponding to eight countable edges per period.

For evaluating these outputs, a hysteresis may be desired. Configuration parameter HYS activates a hysteresis of 45°. When active, the synchronization bits of the serial data output also feature a 45° hysteresis.

HYS Addr. 0x05; bit 1	
Code	Mode
0	No angle hysteresis
1	45° hysteresis on direction change

Table 25: Angle hysteresis

MULTITURN COUNTER

In battery buffered serial interface mode (SEL = low) or I²C slave mode, and as long as the system is powered up correctly (i.e. via battery or main supply), iC-PVL will count the multiturn position. Note that there is no counter overflow handling (positive or negative direction).

The internal counter (MT_COUNT) is 40 bits wide and thus can count up to $2^{40} - 1$ revolutions. In SSI mode, the output bit width is defined by MT_BW, so that $2^{MT_BW} - 1$ revolutions can be counted. Position read-out via SSI or I²C is exclusive, please refer to I²C SLAVE MODE on Page 23.

The counter can be preloaded to a position defined by configuration parameter MT_PREL (Table 26). Refer to Table 19 for the configuration of the counter bit width.

The multiturn counter value as well as the configuration RAM are secured by an eight bit CRC. Refer to chapter I²C MULTIMASTER INTERFACE AND CRC PROTECTION for details.

MT_PREL Addr. 0x06 - 0x0A;	
Code	Value
0x000000000	0
0x000000001	1
...	...
0x0000000FF	255
...	...
0xFFFFFFFFF	$2^{40} - 1$

Table 26: Multiturn preload value

PARALLEL ENCODER MODE (SEL = HIGH)

The input/output signals in parallel encoder mode are described in Figure 11. A start pulse on the PRE line triggers the Hall sensor signal acquisition. The current position is sent as a three bit complementary word via pins P0, N0 to P2, N2. In this mode, the iC-PVL oper-

ates with a single power supply on pin VBAT. Pin VDD must be tied to GND, and the select input SEL must be connected to a logic high level, e.g. VBAT (see circuit in Figure 5).

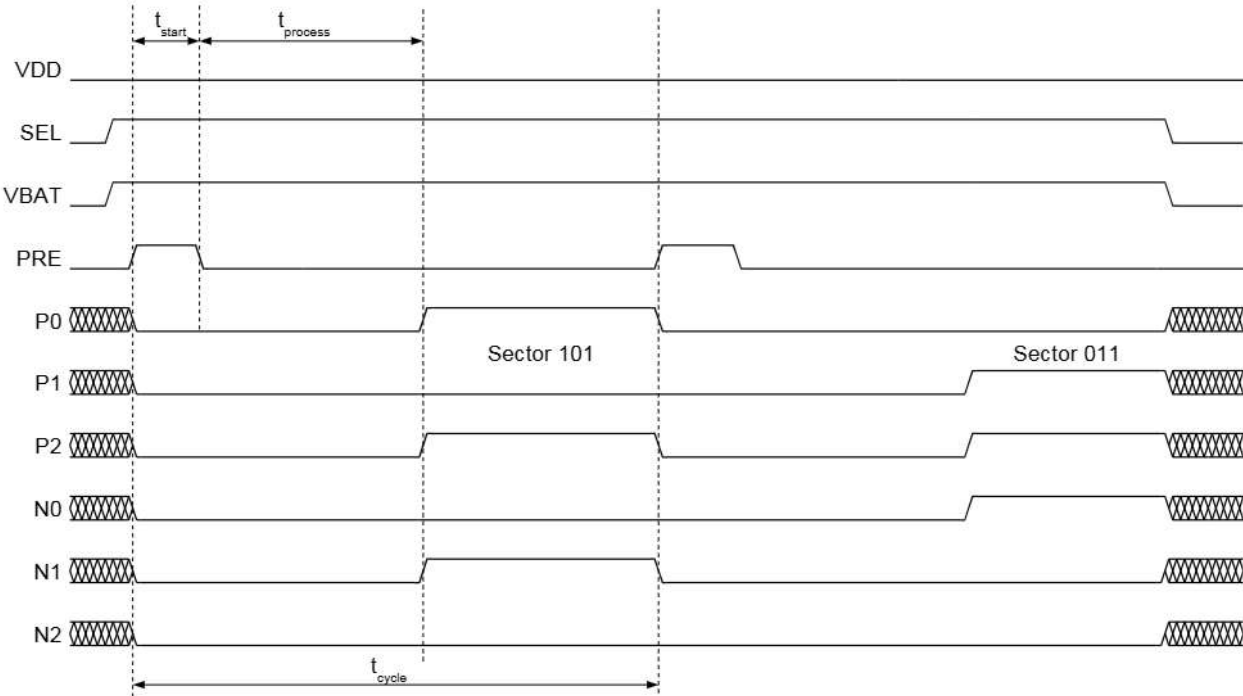


Figure 11: Line signals for parallel encoder mode (3 bit complementary P0-P2 and N0-N2)

I²C MULTIMASTER INTERFACE AND CRC PROTECTION

Pins SCL and SDA form an interface to read an external EEPROM according to the I²C protocol (with at least 128 bytes, e.g. 24C01, 24C02, 24C08 and maximum 24C16, extended address range is not supported). Writing to the EEPROM is not supported.

By default, this EEPROM is used to store the iC-PVL configuration (at addresses 0x00 to 0x0C) according to the register map on Page 10. The configuration is protected against bit errors by an 8-bit CRC checksum. A checksum failure is displayed at output NERR and via the error bit at the end of the SSI data. The multiturn counter preload value is stored in its own configuration area (0x07 - 0x0B) and is also saved with its own CRC on 0x0C. The CRC of the remaining four configuration bytes (0x00 - 0x05) is stored at address 0x06. Both CRC checksums are generated with the polynomial $X^8 + X^5 + X^3 + X^2 + X^1 + 1$ (0x2F, also named 0x12F sometimes). The CRC start value is zero.

Note: In order to avoid an EEPROM content of all bytes = 0x00 to be a valid configuration, the CRC checksums in addresses 0x06 and 0x0C are stored inverted.

Since iC-PVL does only read configuration data, writing EEPROM requires an external programming via pins SCL and SDA (I²C protocol). Refer to circuit on Page 29. In applications with a shared EEPROM, e.g. with iC-MU or iC-MHM, the EEPROM programming of the

iC-PVL configuration can be done via the BiSS interface of the singleturn IC.

If no EEPROM is available or desired in the application, programming the iC-PVL by a microcontroller (MCU) is possible. As described in the subsequent chapter, the I²C slave mode allows direct read/write access to internal configuration and counter. Alternatively, the MCU may emulate an EEPROM (i.e. an I²C slave), since iC-PVL is acting as a bus master by default. At startup, after a short high pulse at pin PRE, the iC-PVL requests addresses 0x00 to 0x0C from the connected I²C slave. This is done in a combined write/read command as shown in Figure 12, repeating 13 times.

The expected slave address here is 0xA0 or "0b 1010 000", the standard I²C EEPROM address.

Notes: In typical applications, the iC-PVL is used in combination with external encoder, line driver or safety ICs. If several devices try to share one common EEPROM, the default configuration area of iC-PVL may not be usable (addresses 0x00 to 0x0C). Therefore, the iC-PVL is capable to **boot from different addresses**. The EEPROM is scanned for the unique iC-PVL configuration footprint, i.e. 13 bytes with correct checksums of configuration and counter preload. If no configuration is found at address **0x00 to 0x0C**, the iC-PVL searches at address **0x40 to 0x4C**, then at address **0x80 to 0x8C** and finally at address **0xA0 to 0xAC**.

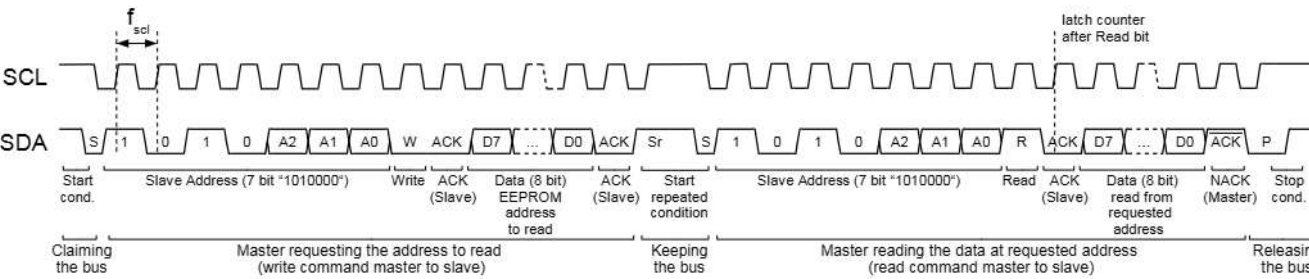


Figure 12: iC-PVL combined write/read command reading one slave address

I²C SLAVE MODE

Additionally to the I²C master interface described in the previous section, iC-PVL can be booted as I²C slave (see circuit of Figure 4 on page 12).
At I²C device ID = 0b1100 001 iC-PVL's internal registers are addressed according to the register map on page 11.

The I²C communication protocol described in Figure 12 may be used for read and write register access. The internal multiturn counter is readable at address 0x07 to 0x0B, protected with CRC checksum at address 0x0C. Additionally, the three synchronization bits are readable at address 0x0D.

Note: For a consistent position information, it is necessary to read all these position registers in one burst command. The counter and synchronization bits are latched after each read addressing (i.e. read/not write bit = high). At burst command, the addressing for read is done only once, for the first address 0x07. The effective clock edge is marked in Figure 12.

If the seven registers are read in seven separate read instructions, the transmitted position may change during readout time and the transmitted position will be inconsistent.

Access to the internal counter via I²C needs to be enabled with configuration parameter I2C_POS. This bit locks position read-out to SSI or I²C exclusively.

I2C_POS Addr. 0x05; bit 7	
Code	Function
0	SSI read-out of MT counter only
1	I ² C read-out of MT counter only

Table 27: Enable I²C or SSI position read-out

For chip release verification purposes an identification value is stored under ROM address 0x0F; a write access to this address is not permitted.

CHIP_REL Adr 0x0F, bit 7:0 (ROM)	
Code	Chip Release
0x01	iC-PVL Y
0x02	iC-PVL Y1
0x03	iC-PVL X
0x04	iC-PVL X1
Note	For all previous versions, address 0x0F is not readable via I2C or answer is 0x00.

Table 28: Chip Release

ERROR MONITOR, STATUS AND COMMAND REGISTER

The iC-PVL has several error conditions. These are stored in a status byte which is readable at I²C address 0x10.

STUP_ERR: Startup Error
Erroneous startup procedure, e.g., I²C stuck-at, EEPROM read error or invalid CRC checksum stored in the EEPROM. No position acquisition is performed. Interfaces are blocked. Please revise configuration and checksums or replace EEPROM. Reboot iC-PVL.

CFG_ERR: Internal Configuration Error
The configuration stored in the internal RAM had an unexpected level flip of one or more bits, visible as wrong CRC checksum. Position is invalid. Reboot iC-PVL.

CTR_ERR: Internal Counter Error
The internal period counter had an unexpected level flip of one or more bits, visible as wrong CRC checksum. Position is invalid. Reboot iC-PVL.

POS_ERR: Position Error
The position encoding observed an unexpected posi-

tion jump, caused e.g. by excessive speed or excessive acceleration of the magnetic disc or tape. Alternatively, this error bit is set on weak, disturbed magnetic signals or complete loss of magnet. Position is invalid. Optimize magnet position and cross-check angular velocity/acceleration with Table 33. Reboot iC-PVL.

BAT_ERR: Battery Error
Battery undervoltage according to Elec. Char. No. 404. Position is invalid. Change battery. Reboot iC-PVL. Battery monitoring is active during VDD supply.

BAT_WRN: Battery Early Warning
Battery voltage early warning according to Elec. Char. No. 405. Battery may be changed during main supply (VDD) as soon as possible. Alternatively, halt system, read current position and restore it. Switch off system, change battery and restart. Restored position may be set as counter preload. Battery monitoring is active during VDD supply.

PDR: Power Down Reset Detected
A power down reset was performed, caused by un-

dervoltage considerably lower than the battery error threshold. Another reason may be due to insufficient stabilizing capacitors at the supply lines VDD and VBAT.

PRESET: Pin Preset Detected

As described on Page 14, a preset pulse at pin PRE is mandatory to initialize and boot-up the iC-PVL after battery insertion. Therefore, this bit is set after each startup. It may be reset by overwriting it or issuing a reset instruction. This can be interpreted as an acknowledgment that this startup was intentional. If the bit is set later during field operation, it is most likely unintentional. The cause may have been an external disruption or short circuit error on the PRE line.

Error Output NERR

An LED may be connected to the error output NERR to signalize errors. The pin is an open drain output driver. If an error is detected, the pin is pulled low. STUP_ERR, CFG_ERR, CTR_ERR, POS_ERR, BAT_ERR are visible at NERR.

Error Output During Startup

During startup phase, a low level at NERR is visible. I.e., from preset pulse until complete and correct configuration read-in from the EEPROM. This indicates that iC-PVL is not ready to operate yet and does not answer a position read request. A reader must wait until error indication is cleared after successful boot-up.

Warning Output NWRN

Battery early warning BAT_WRN is exclusively output at open drain pin NWRN.

Status Byte

Table 29 gives a summary of available error and status messages. Errors can be acknowledged by overwriting the status byte at the desired position or by sending the SCLR command. If the error is still active, iC-PVL error monitor will set it again immediately, i.e, the error condition is still present.

STATUS Addr. 0x10; bit 7:0		
Bit	Name	Description
0	STUP_ERR	Startup error
1	CFG_ERR	Internal configuration error
2	CTR_ERR	Internal counter error
3	POS_ERR	Position error
4	BAT_ERR	Battery error
5	BAT_WRN	Battery early warning
6	PDR	Power down reset detected
7	PRESET	Pin preset detected, I²C REBOOT detected, Sleep mode activated

Table 29: Status byte

Command Register

The **RESET** command reinitializes the internal circuitry. Counter position and configuration remain untouched.

CMD Addr. 0x11; bit 7:0		
Code	Name	Description
0x00	none	Reserved
0x01	none	Reserved
0x02	RESET	Soft reset
0x03	REBOOT	Reboot and preset from EEPROM
0x04	SLEEP	Halt iC-PVL position sensing
0x05	SCLR	Clear all status bits
...	none	No operation

Table 30: Command register

The **REBOOT** command reinitializes the internal circuitry, reloads new configuration and counter preload value from EEPROM. The same actions are performed after a preset pulse at pin PRE.

The **SLEEP** command stops all position sensing action during battery mode and VDD mode. No position is tracked anymore. Power consumption is reduced to a minimum but interfaces are active during VDD supply. This is useful for the storage of encoders with installed battery.

Sleep mode is skipped and position tracking is restarted on any I2C write instruction. Read instructions like status read do not skip sleep mode. With activation of the sleep mode, the status bit 7 (PRESET) is activated and an error is indicated at pin NERR. After leaving the sleep mode, NERR indication is removed and a SCLR command can be used to reset the status bit 7.

The **SCLR** command (Status CLeaR) is used to clear all status messaged in the status register.

SUPPLY SWITCH AND BATTERY MONITORING

To retain and acquire the absolute position even on main power failure, iC-PVL monitors VDD and switches to a battery supply on pin VBAT automatically. The switching point is just below 3 V. So, if the main supply voltage on VDD drops below 3 V, the internal circuitry will be powered by VBAT instead of VDD.

The supply switch features a built-in hysteresis. The threshold voltages are defined in the Electrical Characteristics: V_{off} (Item No. 402), the voltage at which the circuit switches from VDD supply to VBAT supply, and V_{on} (Item No. 401) for the voltage at which the circuit switches back to VDD supply.

BAT_MON		Addr. 0x03; bit 6
Code	Function	
0	Battery monitoring off	
1*	Battery monitoring on	
Note	*) Battery monitoring is active during VDD supply.	

Table 31: Enable battery monitoring

Depending on the power pack, e.g. a 3.6 V battery with 1 Ah capacity, the device can operate for several years. During VDD supply, the iC-PVL monitors the voltage at pin VBAT to detect a low battery voltage. If the supply drops below the error threshold V_{err} voltage (Item No. 404), an error is generated and signaled at

pin NERR, by the error bit in the SSI communication protocol, and by the I²C status register. The battery monitoring function can be enabled/disabled with the configuration parameter BAT_MON (Table 31).

Additionally, if battery monitoring is enabled by parameter BAT_MON, an early battery warning message is generated. The warning threshold is specified in V_{wm} (Elec. Char. No. 405). If the voltage at pin VBAT drops below this warning threshold, a warning is generated and signaled at pin NWRN and optionally by the warning bit of the SSI data (refer to Table 22).

The thresholds of V_{err} and V_{wm} are defined by Elec. Char. 404 and Elec. Char. 405. They can be configured with configuration parameter BAT_THR according to Table 32.

BAT_THR		Addr. 0x05; bit 3:2
Code	V_{err}	V_{wrn}
00*	3.05	3.15
01	2.95	3.05
10	2.85	2.95
11	2.75	2.85
Notes	Nominal values. Refer to Elec. Char. 404 and 405 for variation. *) BAT_THR = 0x00 is recommended.	

Table 32: Battery monitor threshold levels

CURRENT CONSUMPTION IN BATTERY MODE

Serial Interface Mode (SEL = LOW)
In this chapter, all values for angular velocity and acceleration refer to one magnetic period, i.e. one north-south pole pair sequence. One period is 360°. A velocity of one pole pair per second relates to 360°/s = 1 Hz = 60 1/min.

The current consumption of the iC-PVL can be configured by parameter A_MAX. Besides the current consumption, this parameter sets the maximum angle acceleration (from complete halt) supported by the iC-PVL in the respective configuration. The relationship between maximum acceleration and current consumption is shown in Table 33. For accelerations below $48 \cdot 10^3 rad/s^2$, the typical current consumption is below 10 µA.

For applications where only sporadic motor movement is expected during battery supply, Table 33 mainly defines the current consumed by iC-PVL. Additionally, Table 34 gives typical values in case of enduring movement at a certain angular velocity during battery supply.

Note: A magnetic field according to the specifications should be provided any time (see Elec. Char. No. 102). If a suitable magnetic target can not be provided during shipment or storage, use the sleep command (refer to Table 30) or disconnect the battery.

A_MAX Addr. 0x03; bit 5:3				
Code	$\alpha_{max} [\frac{^\circ}{s^2}]$	$\alpha_{max} [\frac{rad}{s^2}]$	typ $I_{avg}[\mu A]$	max $I_{avg}[\mu A]$
000	$160 \cdot 10^6$	$3000 \cdot 10^3$	52	72
001	$40 \cdot 10^6$	$760 \cdot 10^3$	26	36
010	$10 \cdot 10^6$	$190 \cdot 10^3$	14	18
011	$2.5 \cdot 10^6$	$48 \cdot 10^3$	7	10
100	$625 \cdot 10^3$	$12 \cdot 10^3$	4	6
101	$160 \cdot 10^3$	$3 \cdot 10^3$	2.5	4
110	$40 \cdot 10^3$	$0.75 \cdot 10^3$	2	3
111	$10 \cdot 10^3$	$0.2 \cdot 10^3$	1.5	2.5

Table 33: Maximum supported angular acceleration (from shaft halt) and average current consumption on shaft halt or slow angular velocity. V(VBAT) = 3.6 V, V(PRE) < 0.5 V.

I(VBAT) for angular velocity [RPM]								
$f[RPM]$ 2-pole magnet	$f_{mag}[Hz]$ magnetic input freq.	$I_{avg}[\mu A]$ $A_{MAX} = 110$	$I_{avg}[\mu A]$ $A_{MAX} = 101$	$I_{avg}[\mu A]$ $A_{MAX} = 100$	$I_{avg}[\mu A]$ $A_{MAX} = 011$	$I_{avg}[\mu A]$ $A_{MAX} = 010$	$I_{avg}[\mu A]$ $A_{MAX} = 001$	$I_{avg}[\mu A]$ $A_{MAX} = 000$
0	0	2	2.5	4	7	14	26	52
< 125	< 2	2	2.5	4	7	14	26	52
< 250	< 4	2	2.5	4	7	14	26	52
< 500	< 8	2.5	2.5	4	7	14	26	52
< 1000	< 16	4	4	4	7	14	26	52
< 2000	< 33	7	7	7	7	14	26	52
< 4000	< 66	14	14	14	14	14	26	52
< 8000	< 133	26	26	26	26	26	26	52
< 16000	< 266	52	52	52	52	52	52	52
< 32000	< 533	100	100	100	100	100	100	100
< 64000	< 1066	200	200	200	200	200	200	200
< 128000	< 2133	400	400	400	400	400	400	400
< 480000	< 8000	800	800	800	800	800	800	800

Table 34: Average current consumption vs. angular velocity. Typical values for V(VBAT) = 3.6 V, Tj = 27 °C and V(PRE) < 0.5 V.

Parallel Encoder Mode (SEL = HIGH)
The current consumption in parallel encoder mode is directly proportional to the sampling frequency, fs. The typical average current consumption of iC-PVL is calculated as shown below (V(VBAT) = 3.6 V, Tj = 27 °C):

$I_{avg}[\mu A] = 25 \cdot f_s[kHz]$

For instance, at a sampling frequency of 1 000 samples per second: fs = 1 kHz and I = 25 µA. At 100 samples per second: I = 2.5 µA.

OSCILLATOR FREQUENCY CALIBRATION

The bias current for the internal oscillator can be configured with parameter IBIAS. An increase or decrease in bias current will directly affect the oscillator frequency. The bias current should be calibrated at the typical battery supply voltage so that the frequency of the oscillator is around 34 kHz (see Elec. Char. No. 301). The clock frequency is observable at output pin DI_P1 in the dedicated calibration mode. The calibration mode is entered by configuring the EN_ERR parameter to "01" (see Table 21).

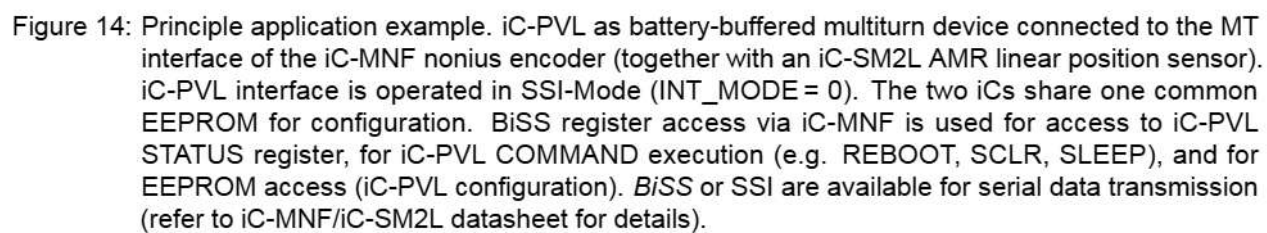
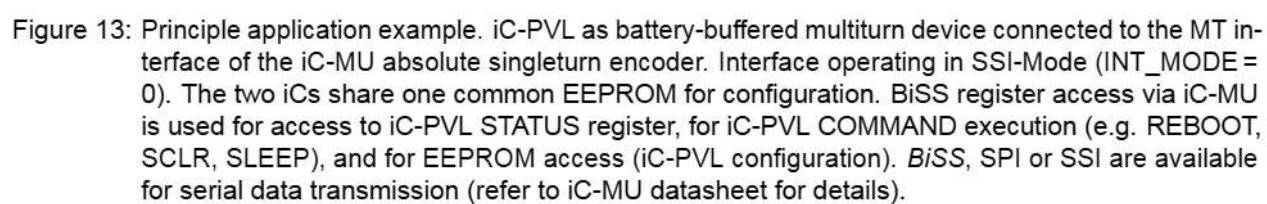
Calibrating the oscillator frequency is not absolutely necessary to ensure iC-PVL operation. Nevertheless, if left uncalibrated, among others, mainly the values defined in Table 33 and 34 may be out of range.

IBIAS		Addr. 0x03; bit 2:0
Code	Frequency change (typ.)	
100	+20 %	
101	+10 %	
110	0 %	
111	-10 %	
000	-20 %	
001	-30 %	
010	-40 %	
011	-50 %	

Table 35: Bias current: Oscillator frequency calibration.

preliminary
ENCODER

APPLICATION EXAMPLES: Singleturn iCs with multiturn interface (SSI Mode)



preliminary
ENCODER

APPLICATION EXAMPLE: iC-PVL as battery powered revolution counter or metering device

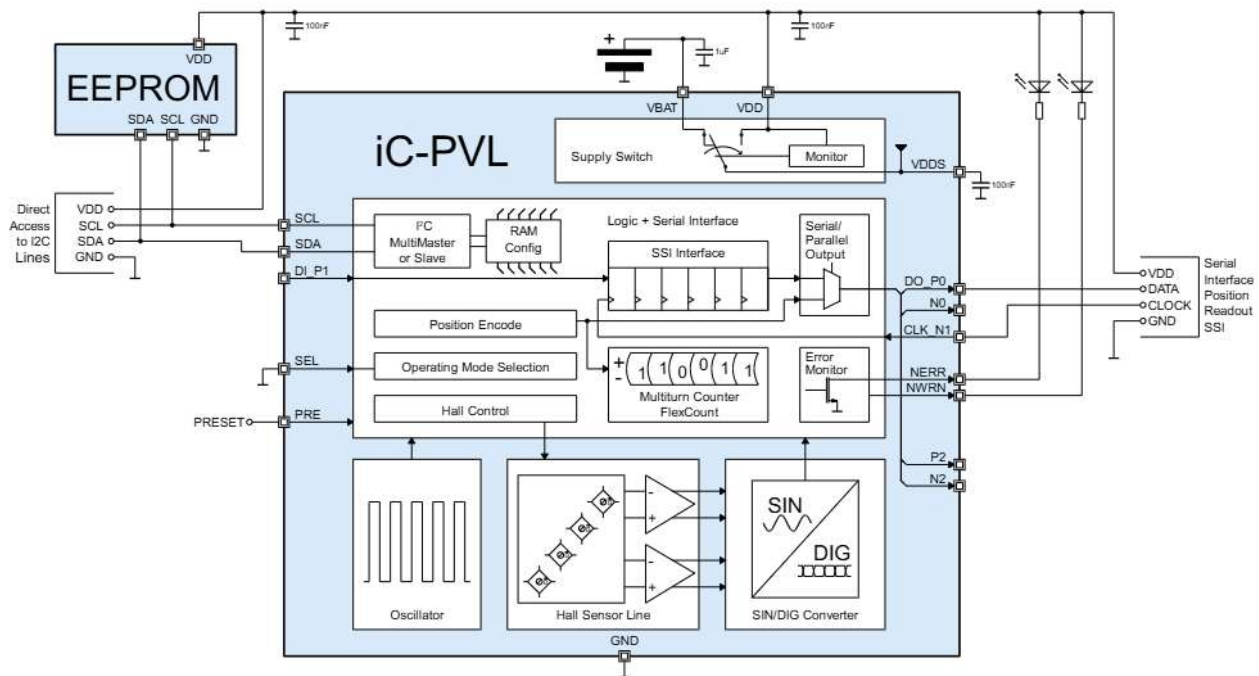


Figure 15: Principle application example. iC-PVL as battery powered multitrans counter, e.g. in a metering application. Interface operating in INT_MODE = 0, read-out of the internal counter value via SSI or I²C. VDD supply is only needed during read-out, otherwise iC-PVL is battery-powered. Direct access to iC-PVL EEPROM via SCL, SDA.

Note: All circuit examples shown in this chapter are principle wiring diagrams. Further components may be necessary but are omitted for clarification of the application principle.

DESIGN REVIEW: Notes On Chip Functions

iC-PVL 0, Y, Y1, X		
No.	Function, Parameter/Code	Description and Application Hints
1		For any former chip release, please refer to datasheet release B1.
2	Battery Assembly and Initial Operation	After installation of the battery, it is important to put iC-PVL into a valid operating state to prevent the battery from an excessive current drain. To do so, a suitable magnetic target needs to be installed, the main supply VDD powered, and a preset pulse applied to input PRE. For details refer to the instructions in chapter STARTUP BEHAVIOUR on P. 14.
3	Magnet Assembly, Shipment and Storage	A magnetic field according to the specifications should be provided any time (see Elec. Char. No. 102). If a suitable magnetic target can not be provided during shipment or storage, use the sleep command (refer to Table 30) or disconnect the battery.

Table 36: Notes on chip functions regarding former iC-PVL chip releases

iC-PVL X1		
No.	Function, Parameter/Code	Description and Application Hints
1	Battery Assembly and Initial Operation	After installation of the battery, it is important to put iC-PVL into a valid operating state to prevent the battery from an excessive current drain. To do so, a suitable magnetic target needs to be installed, the main supply VDD powered, and a preset pulse applied to input PRE. For details refer to the instructions in chapter STARTUP BEHAVIOUR on P. 14.
2	Magnet Assembly, Shipment and Storage	A magnetic field according to the specifications should be provided any time (see Elec. Char. No. 102). If a suitable magnetic target can not be provided during shipment or storage, use the sleep command (refer to Table 30) or disconnect the battery.

Table 37: Notes on chip functions regarding iC-PVL chip release X1

iC-PVL LINEAR/OFF-AXIS
BATTERY-BUFFERED HALL MULTITURN ENCODER

preliminary

Rev B3, Page 31/33

REVISION HISTORY

Rel.	Rel. Date *	Chapter	Modification	Page
A1	2016-06-09	All	Initial release	All

Rel.	Rel. Date *	Chapter	Modification	Page
B1	2016-08-09	ELECTRICAL CHARACTERISTICS	Item 404: max. limits Item 405: max. limits	7
		OPERATING REQUIREMENTS	Figure 1 updated	9
		CONFIGURATION PARAMETERS	Table 5, Table 6: parameter PCR_OUT added	10
		MAGNETIC SCALE SELECTION	Table 11: description of PCR_OUT added	15
		APPLICATION EXAMPLES: Singleturn iCs with multiturn interface (SSI-Mode)	Figure 13 updated	24
		APPLICATION EXAMPLE: iC-PVL as battery powered revolution counter or metering device	Figure 15 updated	25
		DESIGN REVIEW: Notes On Chip Functions	Chapter updated	30

Rel.	Rel. Date *	Chapter	Modification	Page
B2	2017-05-12	ELECTRICAL CHARACTERISTICS	Item 008: Max. tconfig for start adress 0xA0 added	7
		OPERATING MODE SELECTION	Table 8: contents updated	13
		STARTUP BEHAVIOR	Chapter updated Note boxes added	14
		MAGNETIC SCALE SELECTION	Chapter updated Table 9: contents updated	16
		I ² C MULTIMASTER INTERFACE AND CRC PROTECTION	Figure 12: contents updated	22
		I ² C SLAVE MODE	Note box added Table 28 added	23
		ERROR MONITOR, STATUS AND COMMAND REGISTER	Description of sleep mode updated Table 29: contents updated	24
		SUPPLY SWITCH AND BATTERY MONITORING	Table 32: note added	25
		CURRENT CONSUMPTION IN BATTERY MODE	Note box updated	26
		APPLICATION EXAMPLES: Singleturn iCs with multiturn interface (SSI-Mode)	Chapter updated Figure 14 updated	28, 28
		DESIGN REVIEW: Notes On Chip Functions	Chapter updated	30

Rel.	Rel. Date *	Chapter	Modification	Page
B3	2017-08-02	ELECTRICAL CHARACTERISTICS	Item 004: conditions updated	7
		MAGNETIC SCALE SELECTION	Table 9: contents corrected	16
		ERROR MONITOR, STATUS AND COMMAND REGISTER	BAT_ERR, BAT_WRN: description revised	23
		SUPPLY SWITCH AND BATTERY MONITORING	Table 31: note added	25
		CURRENT CONSUMPTION IN BATTERY MODE	Table 34: table heading revised, contents updated Temperature values revised	26

* Release Date format: YYYY-MM-DD

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